

6CCSAHAI

Human AI Interaction

LECTURE 10 – LGT

Red Teaming + evaluation indicators

[What do we want to get done today ?]

1. **Red Teaming Gandalf** [15 min]
2. **Module survey** [5 min]
3. **LGT-task: developing 'indicators of success' for your system** [40 min]
 - Design 3 quantitative & 3 qualitative indicators of your system 'working well' that are measurable
4. **LGT-SGT-task : Let others red-team your system** [3x15 min]
 - Let other people go wild on your system and try to break it, given the indicators of 'success'
5. **SGT task: Write up red-teaming 'report' + potentially fix system** [40 min]

Red teaming

<https://gandalf.lakera.ai/>



You passed level 7.

You completed level 7. Congratulations!

YOU ARE PART OF THE TOP

8%

OF PLAYERS

Secure your spot on the leaderboard by filling in [this form](#).

[MODULE EVALUATION SURVEY]

[LGT task – work in triads]

Your mission: developing ‘indicators of success’ for your system [30 min]

1. Identify the key aspects of how you would know your system succeeded:

What would be happening in participants’ heads? [qualitative]

How would this be visible in the interaction traces? [quantitative]

2. Identify top 2-3 for each of the questions above and decide how you could test these.

3. Present to the rest of the class [2 min per group; ±10 min total]

Petr & Dariana will be coming around to help 1:1 throughout.

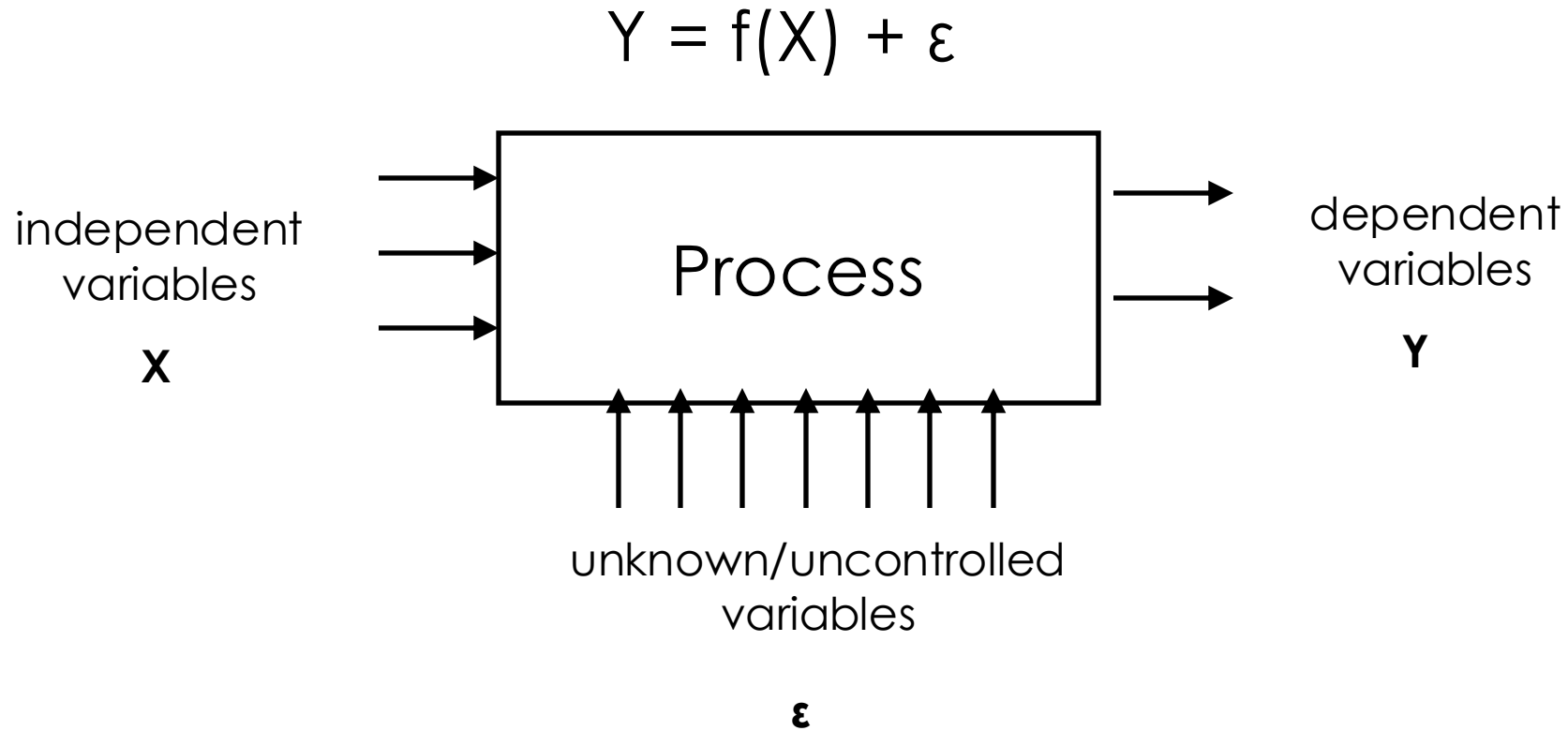
Experiment & Cause

Ex·per·i·ment (ĭk-spĕr'ə-mənt): [Middle English from Old French from Latin *experimentum*, from *experiri*, to try; see *per-* in Indo-European Roots.]
n. Abbr. exp., expt. 1. a. A test under controlled conditions that is made to demonstrate a known truth, examine the validity of a hypothesis, or determine the efficacy of something previously untried. b. The process of conducting such a test; experimentation. 2. An innovative act or procedure: "*Democracy is only an experiment in government*" (William Ralph Inge).

Cause (kôz): [Middle English from Old French from Latin *causa*, reason, purpose.] n. 1. a. The producer of an effect, result, or consequence. b. The one, such as a person, an event, or a condition, that is responsible for an action or a result. v. 1. To be the cause of or reason for; result in. 2. To bring about or compel by authority or force.



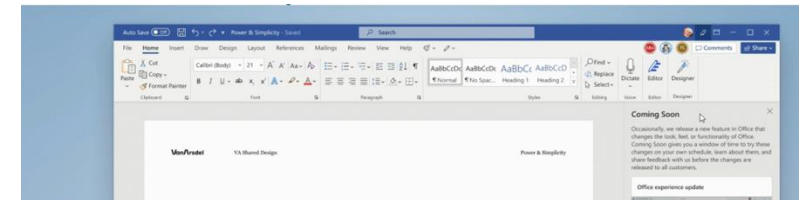
Schematic View of Experiment Design



Controlled Experiment – overview



- Start with a testable **hypothesis**
 - e.g. Mac menu bar is faster than Windows menu bar
- Manipulate **independent variables**
 - different interfaces style (Mac vs Windows)
 - Training given to users on each interface
- Measure **dependent variables**
 - times, errors, # tasks done, satisfaction
- Use **statistical tests** to accept or reject the hypothesis



[LGT task – work in triads]

Your mission: let 3 other groups red-team your system [3x 15 min]

1. Show the red-teamers your list of indicators of success
2. Let them go wild on your system to try to make it do the 'opposite'!
3. Screen capture & record their audio + mark any successful exploit and ask them about it (so you can transcribe afterwards to make your life easier).

[LGT task – work in triads]

Your mission: prepare your report [3x 15 min]

This should include:

1. Evaluation indicators ... you should prepare:

- List of the identified indicators
- Explanation of why you chose each (2-3 sentences)
- How you would measure this in real-world deployment

2. Red teaming report ... you should prepare:

- List of `successful exploits`
- One longer paragraph reflection on what you'd need to change with your system as a result

